37

Candidate Name | Centre Number | Candidate Number

ZIMBABWE SCHOOL EXAMINATIONS COUNCIL

General Certificate of Education Advanced Level

PHYSICS

PAPER 2

9188/2

NOVEMBER 2015 SESSION

1 hour 15 minutes

Candidates answer on the question paper.
Additional materials:
Electronic calculator and/or Mathematical tables

TIME 1 hour 15 minutes

INSTRUCTIONS TO CANDIDATES

FOR EXAMINER'S USE

Write your name, Centre number and candidate number in the spaces at the top of this page.
Answer **all** questions.
Write your answers in the spaces provided on the question paper.
For numerical answers, **all** working should be shown.

INFORMATION FOR CANDIDATES

The number of marks is given in brackets [ ] at the end of each question or part question.

FOR EXAMINATION'S USE	
1	
2	
3	
4	
5	
6	
7	
8	
9	
TOTAL	

This question paper consists of 12 printed pages.

[Turn over

Data

speed of light in free space,	$c = 3.00 \times 10^{8}\ \mathrm{m\,s^{-1}}$
permeability of free space,	$\mu_0 = 4\pi \times 10^{-7}\ \mathrm{H\,m^{-1}}$
permittivity of free space,	$\epsilon_0 = 8.85 \times 10^{-12}\ \mathrm{F\,m^{-1}}$
elementary charge,	$e = 1.60 \times 10^{-19}\ \mathrm{C}$
the Planck constant,	$h = 6.63 \times 10^{-34}\ \mathrm{J\,s}$
unified atomic mass constant,	$u = 1.66 \times 10^{-27}\ \mathrm{kg}$
rest mass of electron,	$m_e = 9.11 \times 10^{-31}\ \mathrm{kg}$
rest mass of proton,	$m_p = 1.67 \times 10^{-27}\ \mathrm{kg}$
molar gas constant,	$R = 8.31\ \mathrm{J\,K^{-1}\,mol^{-1}}$
the Avogadro constant,	$N_A = 6.02 \times 10^{23}\ \mathrm{mol^{-1}}$
the Boltzmann constant,	$k = 1.38 \times 10^{-23}\ \mathrm{J\,K^{-1}}$
gravitational constant,	$G = 6.67 \times 10^{-11}\ \mathrm{N\,m^2\,kg^{-2}}$
acceleration of free fall,	$g = 9.81\ \mathrm{m\,s^{-2}}$

Formulae

uniformly accelerated motion,	$s = ut + \frac{1}{2}at^2$ $v^2 = u^2 + 2as$
work done on/by a gas,	$W = p\Delta V$
gravitational potential,	$\phi = -\frac{Gm}{r}$
refractive index,	$n = \frac{1}{\sin C}$
resistors in series,	$R = R_1 + R_2 + \ldots$
resistors in parallel,	$1/R = 1/R_1 + 1/R_2 + \ldots$
electric potential,	$V = \frac{Q}{4\pi\epsilon_0 r}$
capacitors in series,	$1/C = 1/C_1 + 1/C_2 + \ldots$
capacitors in parallel,	$C = C_1 + C_2 + \ldots$
energy of charged capacitor,	$W = \frac{1}{2}QV$
alternating current/voltage,	$x = x_0 \sin \omega t$
hydrostatic pressure,	$p = \rho gh$
pressure of an ideal gas,	$p = \frac{1}{3}\frac{Nm}{V}\langle c^2 \rangle$
radioactive decay,	$x = x_0 \exp(-\lambda t)$
decay constant,	$\lambda = \frac{0.693}{t_{\frac{1}{2}}}$
critical density of matter in the Universe,	$\rho_0 = \frac{3H_0^2}{8\pi G}$
equation of continuity,	$Av = \text{constant}$
Bernoulli equation (simplified),	$p_1 + \frac{1}{2}\rho v_1^2 = p_2 + \frac{1}{2}\rho v_2^2$
Stokes' law,	$F = Ar\eta v$
Reynolds' number,	$R_e = \frac{\rho v r}{\eta}$
drag force in turbulent flow,	$F = Br^2\rho v^2$

Answer all questions.

For Examiner's Use

1 **(a)** Distinguish between *precision* and *accuracy*.

____

____

____

____ [2]

(b) The speed, V, of ocean waves is given by $V = k\sqrt{g\lambda}$, where k is a constant, λ is the wavelength and g is acceleration of free fall.

Determine the base units for k.

base units for k = ______________________________ [2]

(c) In a simple pendulum experiment to determine the period, T, the equation used is $T = 2\pi\sqrt{\frac{l}{g}}$ where T is found to be (2.16 ± 0.01) s and the length, l, of pendulum is (1.15 ± 0.005) m.

Calculate the value of g and its uncertainty.

g = ____________________ $\pm$ ____________________ [3]

For Examiner's Use

2 (a) Define *acceleration*.

____

____ [1]

(b) A ball of mass 50.0 g is thrown vertically downwards with a velocity of 3.25 ms^{-1}. It hits the ground after 1.62 s.

(i) Assuming air resistance to be negligible, calculate the

1. speed of the ball when it hits the ground,

speed ____________________

2. distance travelled by the ball to the ground.

distance ____________________

(ii) The ball makes contact with the ground for 15 ms and rebounds with an upward velocity of 15.2 ms^{-1}.

Calculate the

1. average force acting on the ball on impact with the ground,

average force ____________________

For Examiner's Use

2. maximum height the ball reaches after hitting the ground.

maximum height ______________________ [8]

3 (i) Explain the origin of the upthrust acting on a body in a fluid.

____

____

____ [2]

(ii) **Fig.3.1** shows a steel ball of diameter 8.2 mm falling through oil at a steady speed.

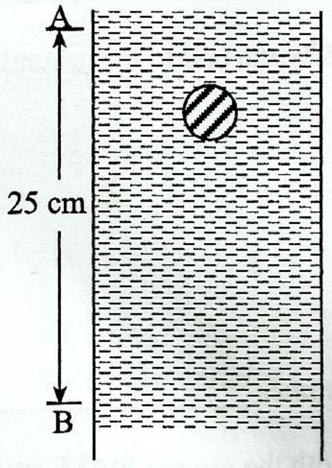

Fig. 3.1

The ball moves from A to B in 0.60 s.

Assuming the density of steel and oil to be 7 800 kgm^{-3} and 900 kgm^{-3} respectively, calculate the

1. weight of the ball,

weight ______________________

For Examiner's Use

2. upthrust on the ball,

upthrust ______________________

3. terminal velocity of the ball.

terminal velocity ______________________ [5]

4 (a) Explain the term *total internal reflection.*

____

____

____ [2]

(b) Give **four** advantages of fibre optic cables over metal cables.

1. ____

____

2. ____

____

3. ____

____

4. ____

____ [4]

For Examiner's Use

5 (a) Define *electric potential.*

____

____ [2]

(b) Equal and opposite $5\,\mu C$ charges are situated at the vertices X and Y of an equilateral triangle XYZ where XY = 10 cm.

Calculate the

(i) magnitude of the electric field strength at Z,

field strength = ________________________

(ii) electric potential at *Z*.

electric potential = ________________________ [4]

(c) Describe how dust can be extracted electrostatically from a chimney of a factory.

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____

____ [3]

For Examiner's Use

6 (a) Describe the principle of operation of an ideal transformer.

____

____

____ [3]

(b) The primary coil of an ideal transformer has 480 turns and is connected to a 120 V (r.m.s), 60 Hz sinusoidal supply. The secondary coil has 40 turns and is connected to a resistor of resistance 4.0 Ω as shown in **Fig. 6.1.**

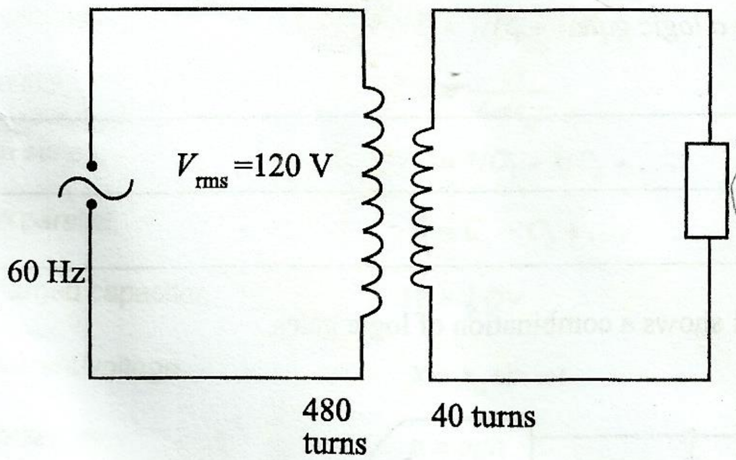

Fig.6.1.

Calculate

(i) V_{rms} across the resistor,

V_{rms} = ______________________

[Turn over

For Examiner's Use

(ii) peak power dissipated by the resistor.

peak power ______________________ [4]

7 (a) Define *a logic gate.*

____

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______________________ [1]

(b) **Fig.7.1** shows a combination of logic gates.

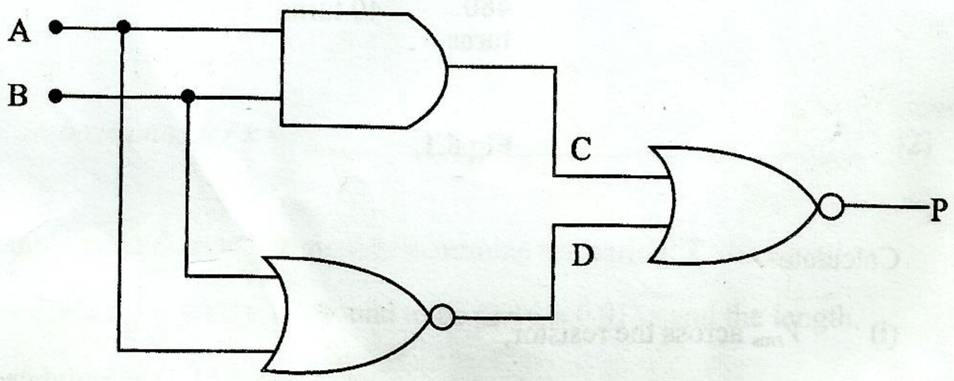

Fig.7.1

For Examiner's Use

(i) Construct a truth table for the combination.

(ii) State the logical function for the combination.

____ [4]

8 Different types of thermometers have different thermometric properties and advantages.

Complete **Table 8.1** to distinguish between the two types of thermometers given.

Table 8.1

type of thermometer	thermometric property	advantages
liquid in glass		
thermocouple		

[4]

For Examiner's Use

9 (a) Radioactive decay is *random* and *spontaneous.*

Define the terms in italics.

spontaneous ______________________________

____

random ______________________________

______________________________ [2]

(b) Radium-226 has a decay constant of 1.8×10^{-6} s and an initial decay rate of $5.2 \times 10^{10}\ s^{-1}$

Calculate the

(i) initial number of radioactive atoms present,

number of atoms = ______________________

(ii) time taken for the activity to fall to a quarter of its initial value.

time = ______________________

[4]